Markdown

Lightweight prose, but ket, bra, and integrals remain embedded TeX; complex structures and numbering stay manual.

```
# Quantum States and Measurement

Consider a two-level system:
- $|0\rangle,|1\rangle$ form an orthonormal
basis;
- $|\psi\rangle=\frac{|0\rangle+i|1\rangle}
{\sqrt2}$.

## Definition 1.1 (State expansion)
$$
|\psi(t)\rangle=\sum_n c_n(t)|n\rangle
$$
$$
i\hbar\frac{\partial}{\partial t}|
\psi(t)\rangle
=\hat H|\psi(t)\rangle
$$
The expectation value of $\hat A$ is
$$
\langle\psi|\hat A|\psi\rangle
=\int\psi^\*(x)\hat A\psi(x)\,\mathrm{d}x
$$
```

```
## Theorem 1.1 (Normalization)
If $\hat H=\hat H^\dagger$, then
$\sum_n|c_n(t)|^2=1$ for every $t$.

> **Proof.** Differentiate Definition 1.1
> and use the Hermiticity of the Hamiltonian.

## Problem 1.1
Measure
$|\psi\rangle=\frac{|0\rangle+i|1\rangle}
{\sqrt2}$.
Find the probabilities of the two basis
states.

**Solution.**
$P(0)=|\langle0|\psi\rangle|^2=\frac12$,
$P(1)=|\langle1|\psi\rangle|^2=\frac12$.

| State | Amplitude | Probability |
|---|---:|---:|
| $|0\rangle$ | $\langle0|\psi\rangle=1/
\sqrt2$ | $1/2$ |
| $|1\rangle$ | $\langle1|\psi\rangle=i/
\sqrt2$ | $1/2$ |

**Table 1.1  Measurement outcomes**
```

Note Manual maintenance

Headings, itemize, and basic tables are direct; block styles, three-line tables, numbering, resets, and cross-references must be kept consistent by hand. Rendering options are limited too.

LaTeX

Fully capable, but itemize, theorem blocks, ket/bra, and three-line tables require explicit commands or packages.

```latex
\usepackage{amsmath,amsthm,braket,booktabs}

\section{Quantum States and Measurement}
Consider a two-level system:
\begin{itemize}
  \item $\ket{0},\ket{1}$ form an orthonormal
basis;
  \item $\ket{\psi}
    =\frac{\ket{0}+i\ket{1}}{\sqrt{2}}$.
\end{itemize}

\begin{definition}[State
expansion]\label{def:state}
  \[
    \ket{\psi(t)}=\sum_n c_n(t)\ket{n}
  \]
  \[
    i\hbar\frac{\partial}{\partial
t}\ket{\psi(t)}
    =\hat H\ket{\psi(t)}
  \]
  The expectation value of $\hat A$ is
  \[
    \bra{\psi}\hat A\ket{\psi}
    =\int\psi^*(x)\hat A\psi(x)\,\mathrm{d}x
  \]
\end{definition}
```

```latex
\begin{theorem}
[Normalization]\label{thm:norm}
  If $\hat H=\hat H^\dagger$, then
  $\sum_n|c_n(t)|^2=1$ for every $t$.
\end{theorem}
\begin{proof}
  Differentiate Definition~\ref{def:state}
  and use the Hermiticity of the Hamiltonian.
\end{proof}

\begin{problem}
  Measure
  $\ket{\psi}=\frac{\ket{0}+i\ket{1}}
{\sqrt{2}}$.
  Find the probabilities of the two basis
states.
\end{problem}
\begin{solution}
  $P(0)=|\braket{0|\psi}|^2=\frac{1}{2}$,
  $P(1)=|\braket{1|\psi}|^2=\frac{1}{2}$.
\end{solution}

\begin{table}
  \centering
  \begin{tabular}{ccc}
    \toprule
    State & Amplitude & Probability \\
    \midrule
    $\ket{0}$ & $\braket{0|\psi}=1/\sqrt2$ &
$1/2$ \\
    $\ket{1}$ & $\braket{1|\psi}=i/\sqrt2$ &
$1/2$ \\
    \bottomrule
  \end{tabular}
  \caption{Measurement outcomes}
\end{table}
```

Note Boilerplate

The semantics are precise and the controls are deep, but you also maintain environments, labels, references, packages, and the build toolchain. Backslash-heavy source is harder to scan, and large local builds can interrupt the writing flow.

Typst + Scripst

Lists, physics notation, and formulas stay compact; countblocks and three-line tables use ready-made styles.

```
= Quantum States and Measurement

Consider a two-level system:

- $ket(0), ket(1)$ form an orthonormal basis;
- $ket(Psi) = (ket(0) + i ket(1))/sqrt(2)$.

#definition(subname: [State expansion], lab:
"def:state")[
  $
    ket(Psi(t)) = sum_n c_n (t) ket(n)
  $
  $
    i hbar dv(, t) ket(Psi(t)) = hat(H)
ket(Psi(t))
  $
  The expectation value of $hat(A)$ is
  $
    braket(Psi, hat(A), Psi)
      = integral Psi^*(x) hat(A) Psi(x) dif x
  $
]
```

```
#theorem(subname: [Normalization], lab:
"thm:norm")[
  If $hat(H) = hat(H)^dagger$, then
  $sum_n abs(c_n (t))^2 = 1$ for every $t$.
]

#proof[
  Differentiate @def:state and use the
  Hermiticity of the Hamiltonian.
]

#problem(lab: "prob:measure")[
  Measure $ket(Psi) = (ket(0) + i ket(1))/
sqrt(2)$.
  Find the probabilities of the two basis
states.
]

#solution[
  $P(0) = abs(braket(0, Psi))^2 = 1/2$,
  $P(1) = abs(braket(1, Psi))^2 = 1/2$.
]

#figure(
  three-line-table[
    | State | Amplitude | Probability |
    | $ket(0)$ | $braket(0, Psi) = 1/sqrt(2)$
| $1/2$ |
    | $ket(1)$ | $braket(1, Psi) = i/sqrt(2)$
| $1/2$ |
  ],
  caption: [Measurement outcomes],
)
```

Note Scripst + Ratchet

Scripst integrates Physica, Tablem, and content-block presets, while Ratchet keeps numbering depth, resets, and references consistent.

Real Scripst Output

These are not imitation boxes: the physics notation, three-line table, and content blocks below are generated by Scripst itself.

Quantum States and Measurement

Consider a two-level system:

- $|0\rangle, |1\rangle$ form an orthonormal basis;
- $|\Psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$.

Definition 1.1 State expansion

$$|\Psi(t)\rangle = \sum_{n} c_n(t)|n\rangle \tag{1.1}$$

and satisfies

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle \tag{1.2}$$

The expectation value of $\hat{A}$ is

$$\left\langle \Psi \middle| \hat{A} \middle| \Psi \right\rangle = \int \Psi^*(x)\hat{A}\Psi(x)\,\mathrm{d}x \tag{1.3}$$

Theorem 1.1 Normalization

If $\hat{H} = \hat{H}^\dagger$, then $\sum_n |c_n(t)|^2 = 1$ for every t.

Proof. Differentiate Definition 1.1 and use the Hermiticity of the Hamiltonian. ∎

Problem 1.1

Measure $|\Psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$ and find the probabilities of the two basis states.

Solution. $P(0) = |\langle 0|\Psi\rangle|^2 = \frac{1}{2}$, and $P(1) = |\langle 1|\Psi\rangle|^2 = \frac{1}{2}$.

State	Amplitude	Probability
$\|0\rangle$	$\langle 0\|\Psi\rangle = \frac{1}{\sqrt{2}}$	$\frac{1}{2}$
$\|1\rangle$	$\langle 1\|\Psi\rangle = \frac{i}{\sqrt{2}}$	$\frac{1}{2}$

Table 1.1: Measurement outcomes